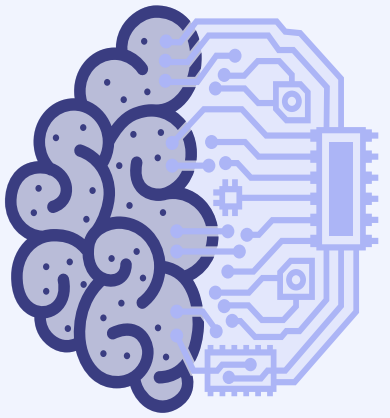




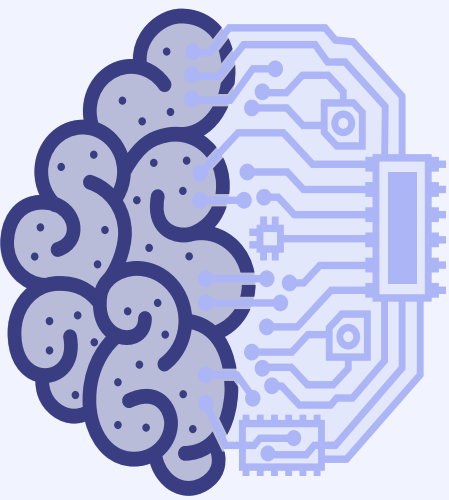
SLEEP STAGES CLASSIFICATION

USING DEEP LEARNING

supervised by :

DR. Ibrahim Sadek

ENG. Veronica William

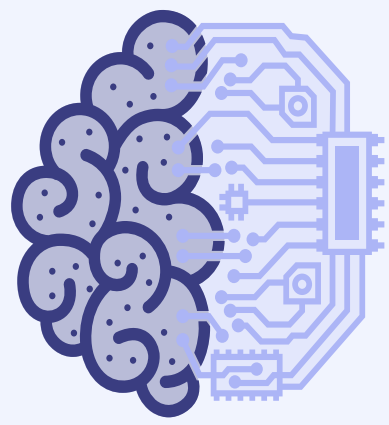


OUR TEAM

WAFAA HASSAN RAMADAN

SOHAILA ELSAYED IBRAHIM

MENNAT ALLAH MOHAMED ELSAYED



INTRODUCTION

Develop a deep learning-based model for automatic sleep stage classification from physiological signals.

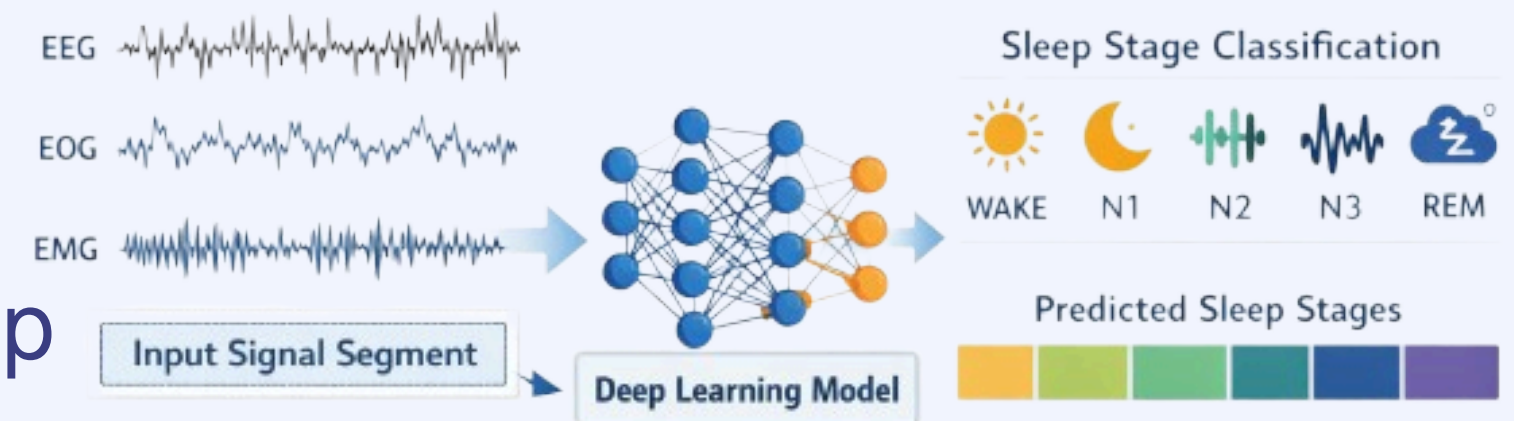
The model analyzes time-series signal data and classifies each segment into one of the five sleep stages:

Wake, N1, N2, N3, and REM.

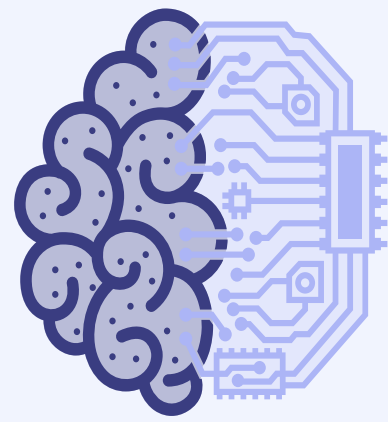
The proposed system aims to improve the efficiency and accuracy of sleep analysis.

It can assist clinicians by reducing manual scoring effort and increasing the reliability of sleep stage detection.

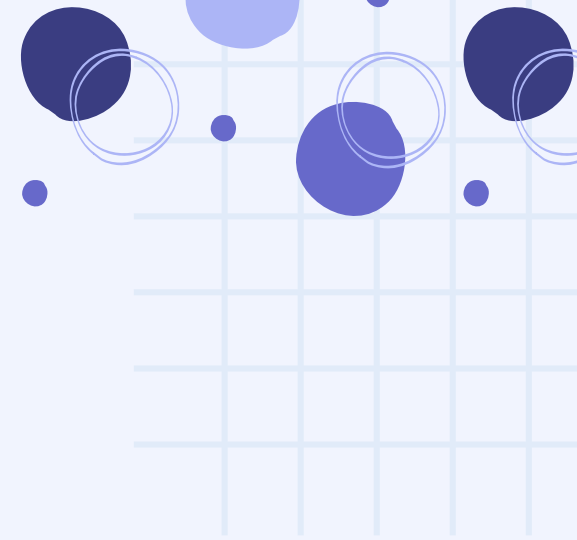
Automatic Sleep Stage Classification



Improving Sleep Analysis with AI
Accurate & Efficient Sleep Stage Detection



DATASET



The MESA (Multi-Ethnic Study of Atherosclerosis) Sleep Dataset is a large-scale, comprehensive collection of physiological sleep data.

It is hosted by the National Sleep Research Resource (NSRR) and was originally designed to study the links between sleep characteristics and cardiovascular health across a diverse demographic of over 2,000 participants.

Three Pillars of Sleep Data Collection



Polysomnography (PSG)

- Full overnight unattended recordings.
- Raw data files available.



Actigraphy

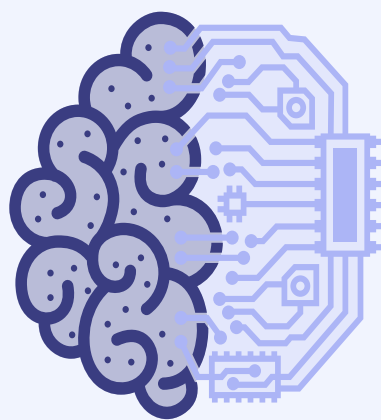
- 7-day wrist-worn monitoring.
- Raw data files available.



Questionnaires

- Self-reported sleep questionnaires mapped to key sets of MESA Sleep covariates.

©: NotebookLM

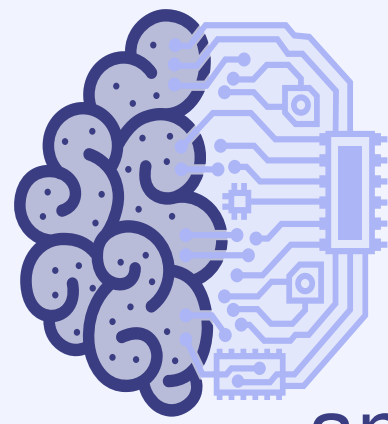


DATASET

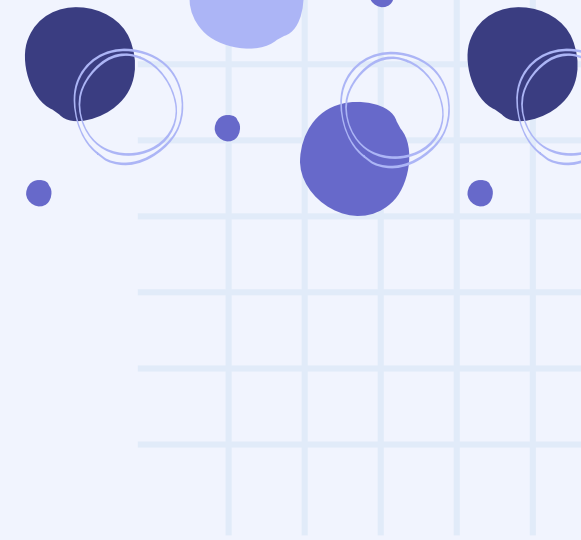
Dataset Structure:

Signal Data (X):

- The input data consists of time-series physiological signals. These signals provide valuable information about brain activity and physiological changes occurring during different sleep stages.
- Each signal sample is represented as a sequence of numerical values describing signal amplitude over time. These signals include measurements of:
[EKG, EOG-L, EOG-R, EMG, EEG1, EEG2, EEG3, Pres, Flow, Snore, Thor, Abdo, Leg, Therm, Pos, EKG_Off, EOG-L_Off, EOG-R_Off, EMG_Off, EEG1_Off, EEG2_Off, EEG3_Off, Pleth, OxStatus, SpO2, HR, DHR]



DATASET



annotation (labels)_ (y).

The XML file acts as a "Time-Stamped Log" or "Metadata Map" created by sleep technicians to describe the continuous signals found in the EDF file.

Sleep Staging :Divides the entire recording into 30-second epochs.

Scored Events (Clinical Observations):Respiratory Events - Oxygen Desaturation - Limb Movements

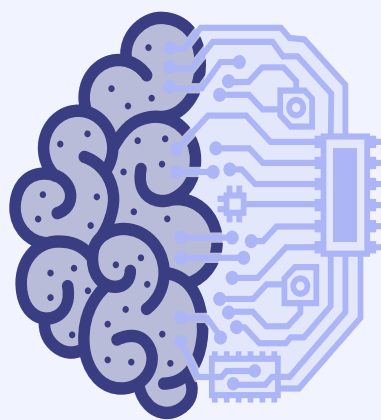
Sleep Stage	Description
Wake	The subject is awake
N1	Light sleep stage
N2	Intermediate sleep stage
N3	Deep sleep stage
REM	Rapid Eye Movement sleep

ELECTROOCULOGRAPHY (EOG) SIGNAL

- **EOG:** Non-invasive electrophysiological technique that records corneo-retinal standing potential
- Captures horizontal and vertical eye movements through potential differences between electrodes

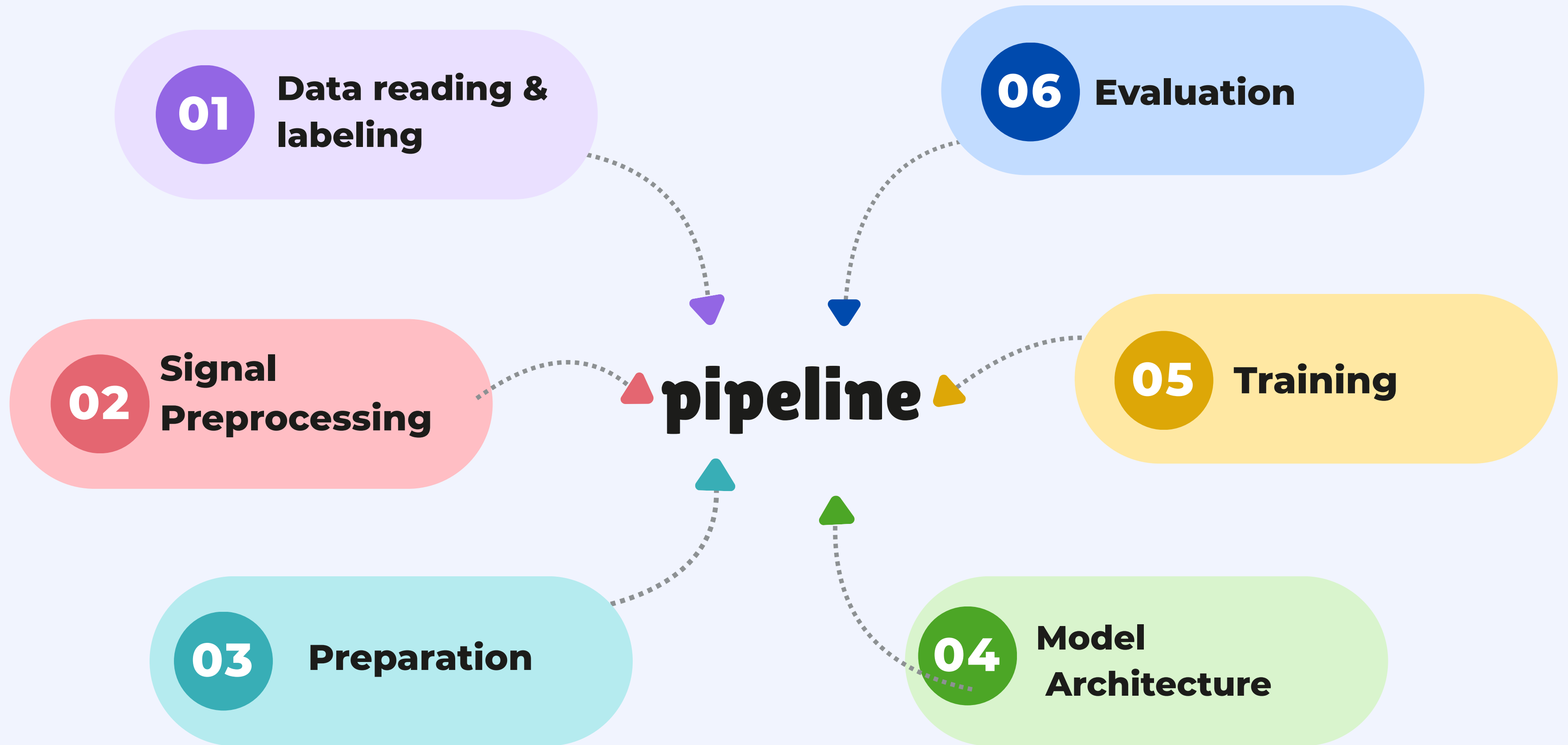
Key Advantages:

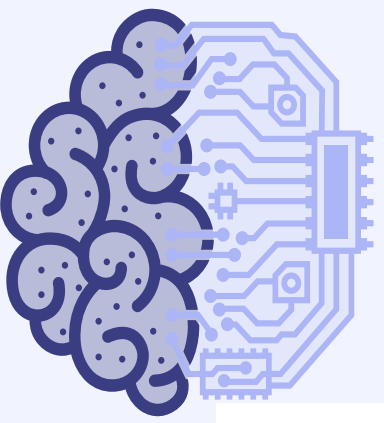
- Less invasive than EEG (fewer electrodes)
- Strong correlation with REM sleep (eye movements)
- More suitable for wearable devices
- Lower cost and easier setup



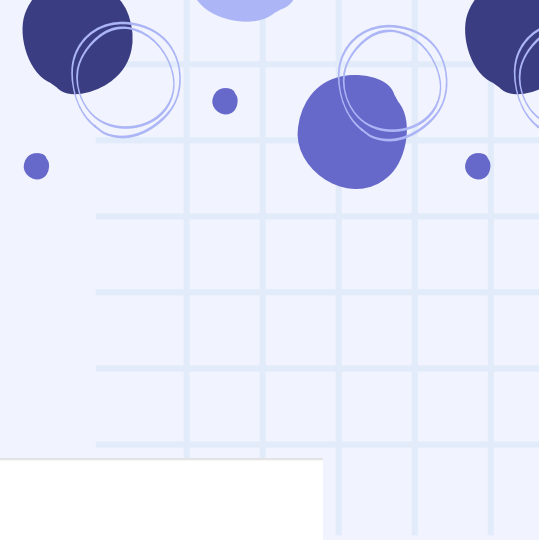
PROJECT OBJECTIVES

- Develop a robust deep learning framework for multi-class sleep stage classification (Wake, N1, N2, N3, REM) using single-channel EOG
- Transition from conventional multi-channel EEG toward scalable, wearable, and minimally intrusive solutions
- Address inherent class imbalance in sleep stage distribution through advanced sampling and weighting strategies
- Achieve competitive performance (~80% test accuracy) with a computationally lightweight architecture suitable for edge devices





DATA READING



```
Number of individuals present: 10
Folder names: ['0001', '0002', '0006', '0010', '0012', '0014', '0016', '0021', '0027', '0028']

→ Folder 0001:
  EDF files: 1 → ['mesa-sleep-0001.edf']
  XML files: 1 → ['mesa-sleep-0001-nsrr.xml']

→ Folder 0002:
  EDF files: 1 → ['mesa-sleep-0002.edf']
  XML files: 1 → ['mesa-sleep-0002-nsrr.xml']

→ Folder 0006:
  EDF files: 1 → ['mesa-sleep-0006.edf']
  XML files: 1 → ['mesa-sleep-0006-nsrr.xml']

→ Folder 0010:
  EDF files: 1 → ['mesa-sleep-0010.edf']
  XML files: 1 → ['mesa-sleep-0010-nsrr.xml']

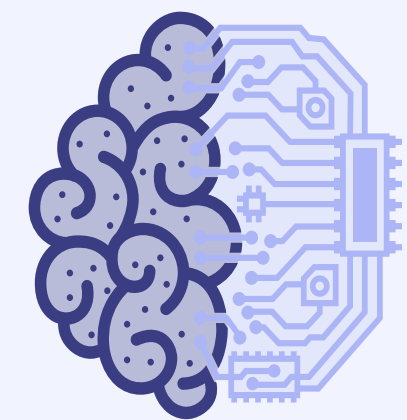
→ Folder 0012:
  EDF files: 1 → ['mesa-sleep-0012.edf']
  XML files: 1 → ['mesa-sleep-0012-nsrr.xml']

→ Folder 0014:
  EDF files: 1 → ['mesa-sleep-0014.edf']
  XML files: 1 → ['mesa-sleep-0014-nsrr.xml']

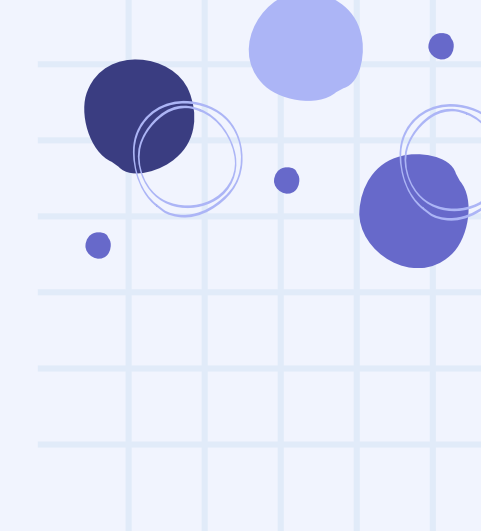
→ Folder 0016:
  EDF files: 1 → ['mesa-sleep-0016.edf']
  XML files: 1 → ['mesa-sleep-0016-nsrr.xml']
```

Channel Names for Subject 0001

1. EKG
2. EOG-L
3. EOG-R
4. EMG
5. EEG1
6. EEG2
7. EEG3
8. Pres
9. Flow
10. Snore
11. Thor
12. Abdo
13. Leg
14. Therm
15. Pos
16. EKG_off
17. EOG-L_off
18. EOG-R_off
19. EMG_off
20. EEG1_off
21. EEG2_off
22. EEG3_off
23. Pleth
24. OxStatus
25. SpO2
26. HR
27. DHR



LABELING



```
Subject: 0001
Total Epochs: 1361
Stages Summary -> Wake: 752 | N1: 135 | N2: 455 | N3: 19 | REM: 0
Subject: 0002
Total Epochs: 1138
Stages Summary -> Wake: 571 | N1: 49 | N2: 362 | N3: 156 | REM: 0
Subject: 0006
Total Epochs: 960
Stages Summary -> Wake: 364 | N1: 141 | N2: 380 | N3: 75 | REM: 0
Subject: 0010
Total Epochs: 1199
Stages Summary -> Wake: 980 | N1: 20 | N2: 126 | N3: 73 | REM: 0
Subject: 0012
Total Epochs: 1359
Stages Summary -> Wake: 882 | N1: 85 | N2: 376 | N3: 16 | REM: 0
Subject: 0014
Total Epochs: 1501
Stages Summary -> Wake: 836 | N1: 22 | N2: 381 | N3: 262 | REM: 0
Subject: 0016
Total Epochs: 977
Stages Summary -> Wake: 355 | N1: 72 | N2: 511 | N3: 39 | REM: 0
Subject: 0021
Total Epochs: 1002
Stages Summary -> Wake: 139 | N1: 218 | N2: 512 | N3: 133 | REM: 0
Subject: 0027
```

SIGNAL PREPROCESSING



FILTERING:

- Band-pass filter (0.3 – 30 Hz) applied to EOG signals.



NORMALIZATION:

- Z-score normalization
- Ensures stable training across different participants



CHANNEL SELECTION:

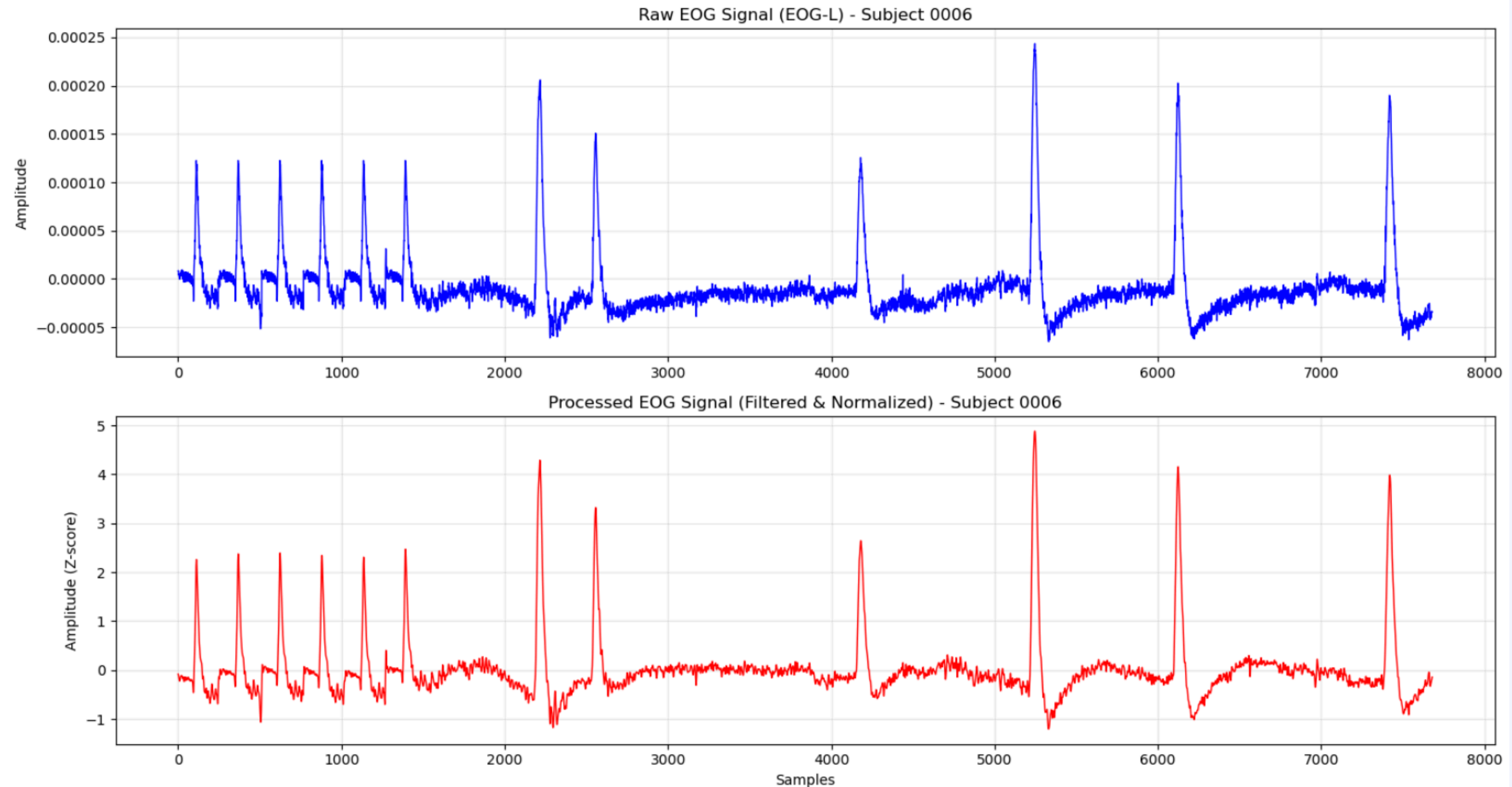
- Left EOG channel



EPOCHING:

- into 30 seconds to match and align the annotations, given a sampling rate of 256 Hz, each epoch contains 30×256 data points

RAW VS PROCESSED.



RAW VS PROCESSED.

```
*** Raw Signal Length: 8294144 samples
Processed File Loaded - Number of epochs: 1079
```

```
--- Raw vs Processed - Subject 0006 (First Epoch) ---
```

Metric	Raw Signal	Processed Signal	Diff
Mean	-0.000009	-0.002399	-0.002390
Std	0.000035	0.667366	0.667331
Min	-0.000065	-1.204099	-1.204034
Max	0.000244	4.882393	4.882149
Range	0.000309	6.086492	

```
First 15 values:
```

Sample 0	Raw: 0.00001	Processed: -0.08014	Diff: -0.08014
Sample 1	Raw: 0.00001	Processed: -0.09835	Diff: -0.09835
Sample 2	Raw: 0.00001	Processed: -0.11717	Diff: -0.11717
Sample 3	Raw: 0.00000	Processed: -0.13701	Diff: -0.13702
Sample 4	Raw: 0.00000	Processed: -0.15782	Diff: -0.15782
Sample 5	Raw: 0.00000	Processed: -0.17868	Diff: -0.17869
Sample 6	Raw: 0.00000	Processed: -0.19773	Diff: -0.19774
Sample 7	Raw: 0.00000	Processed: -0.21251	Diff: -0.21251
Sample 8	Raw: -0.00000	Processed: -0.22084	Diff: -0.22084
Sample 9	Raw: 0.00000	Processed: -0.22174	Diff: -0.22174
Sample 10	Raw: 0.00000	Processed: -0.21554	Diff: -0.21554
Sample 11	Raw: -0.00000	Processed: -0.20353	Diff: -0.20353
Sample 12	Raw: 0.00000	Processed: -0.18751	Diff: -0.18751
Sample 13	Raw: 0.00001	Processed: -0.16951	Diff: -0.16952
Sample 14	Raw: 0.00000	Processed: -0.15166	Diff: -0.15166

DATA PREPARATION & SPLITTING

DATA ALIGNMENT: Signals are divided into 30-second epochs (7680 samples at 256 Hz)
Each epoch is matched with its sleep stage label from the XML file
Invalid epochs or those containing artifacts are discarded

LABEL ENCODING: Sleep stage labels are converted to numerical values: 0 = Wake, 1 = N1, 2 = N2, 3 = N3, 4 = REM

HANDLING IMBALANCE: Sleep stages are highly imbalanced (especially N1 and N3)
Techniques Applied: SMOTE + Class Weights
This improves model performance on minority classes

DATA SPLIT: The dataset is divided into:

- 70% Training
- 15% Validation
- 15% Test

MODEL ARCHITECTURE

-Hybrid Model (CNN + LSTM + Attention)

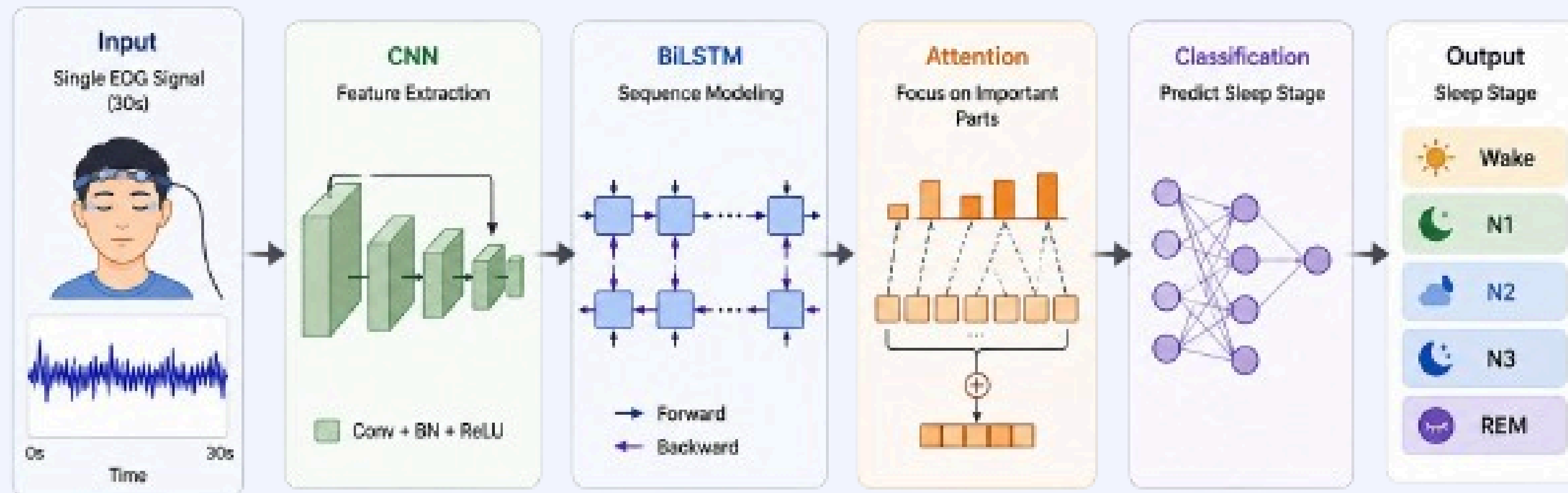
-Input Single EOG Signal (30 seconds)

-Model Structure:

- Residual CNN Extracts important features from the signal
- Bidirectional LSTM Understands the sequence and timing of sleep patterns
- Attention Mechanism Focuses on the most relevant parts of the signal
- Classification Layer Predicts one of 5 sleep stages:
Wake | N1 | N2 | N3 | REM

Goal: High accuracy with a lightweight model suitable for wearable devices

MODEL ARCHITECTURE



MODEL EVOLUTION ACROSS NOTEBOOKS

We developed **4 versions** of the project to improve performance step by step:

Notebook	Data Splitting	Imbalance Handling	Augmentation	Test Accuracy
signals_eog1	Subject-wise	WeightedRandomSampler	Noise + Scaling	65.95%
signalseog2	Sampling	WeightedRandomSampler	Noise + Scaling + Time Shifting	74.50%
signalseog3	Subject-wise	SMOTE	Noise + Scaling	80.25%
signalseog4 (Final)	Sampling	SMOTE	Noise + Scaling	82.39%

TRAINING PROCESS & RESULTS

-Training Parameters:

- Epochs: 50
- Batch Size: 32

-Key Observations During Training:

- Training Accuracy reached **99.5%** by the final epoch
- Best Validation Accuracy: **~83.2%**
- Model shows good convergence with stable training

TRAINING RESULTS

1ST VERSION "SIGNALS_EOG1"

```
Train Loss: 0.4366 | Train Acc: 0.9542 | Val Loss: 1.3961 | Val Acc: 0.7751  
Epoch 50/50 [Train]: 100%|██████████| 267/267 [00:24<00:00, 11.04it/s, Loss=0.408, LR=4.03e-9]  
Train Loss: 0.4363 | Train Acc: 0.9560 | Val Loss: 1.4021 | Val Acc: 0.7722
```

2ND VERSION "SIGNALS_EOG2"

```
Train Loss: 0.1615 | Train Acc: 0.9013 | Val Loss: 1.0154 | Val Acc: 0.7529  
No improvement. (Count: 14/15)  
Epoch 47/50 [Train]: 100%|██████████| 279/279 [00:30<00:00, 9.14it/s, Loss=0.195, LR=1.8e-5]  
Train Loss: 0.1521 | Train Acc: 0.9092 | Val Loss: 1.1881 | Val Acc: 0.7257
```

3RD VERSION "SIGNALS_EOG3"

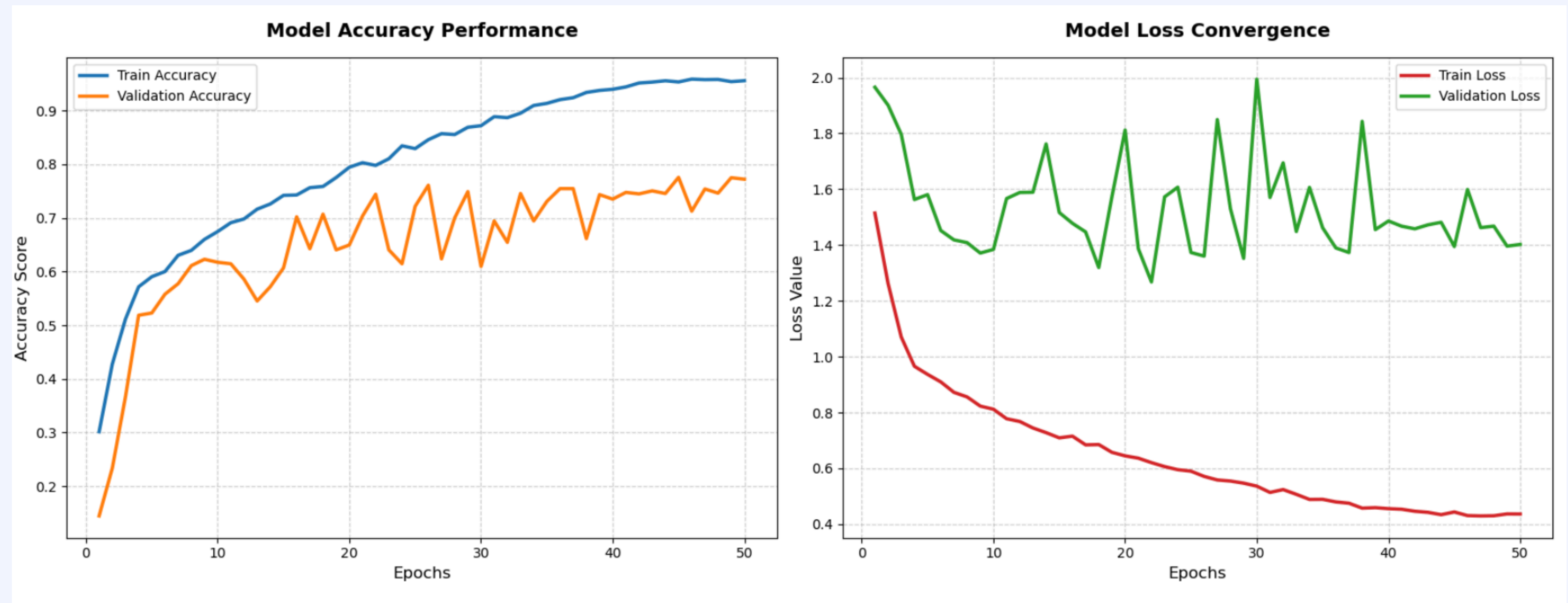
```
Train Loss: 0.4378 | Train Acc: 0.9949 | Val Loss: 0.8523 | Val Acc: 0.8130  
Epoch 50/50 [Train]: 100%|██████████| 609/609 [00:53<00:00, 11.42it/s, Loss=0.436, LR=4.01e-9]  
Train Loss: 0.4384 | Train Acc: 0.9948 | Val Loss: 0.9230 | Val Acc: 0.7915  
Training completed!
```

4TH VERSION "SIGNALS_EOG4"

```
Epoch 47/50 [Train]: 100%|██████████| 609/609 [00:50<00:00, 10.50it/s, Loss=0.42, LR=4.01e-9]  
Train Loss: 0.4410 | Train Acc: 0.9940 | Val Loss: 0.8303 | Val Acc: 0.8192  
Epoch 50/50 [Train]: 100%|██████████| 609/609 [01:02<00:00, 9.78it/s, Loss=0.426, LR=4.01e-9]  
Train Loss: 0.4406 | Train Acc: 0.9940 | Val Loss: 0.8041 | Val Acc: 0.8323
```

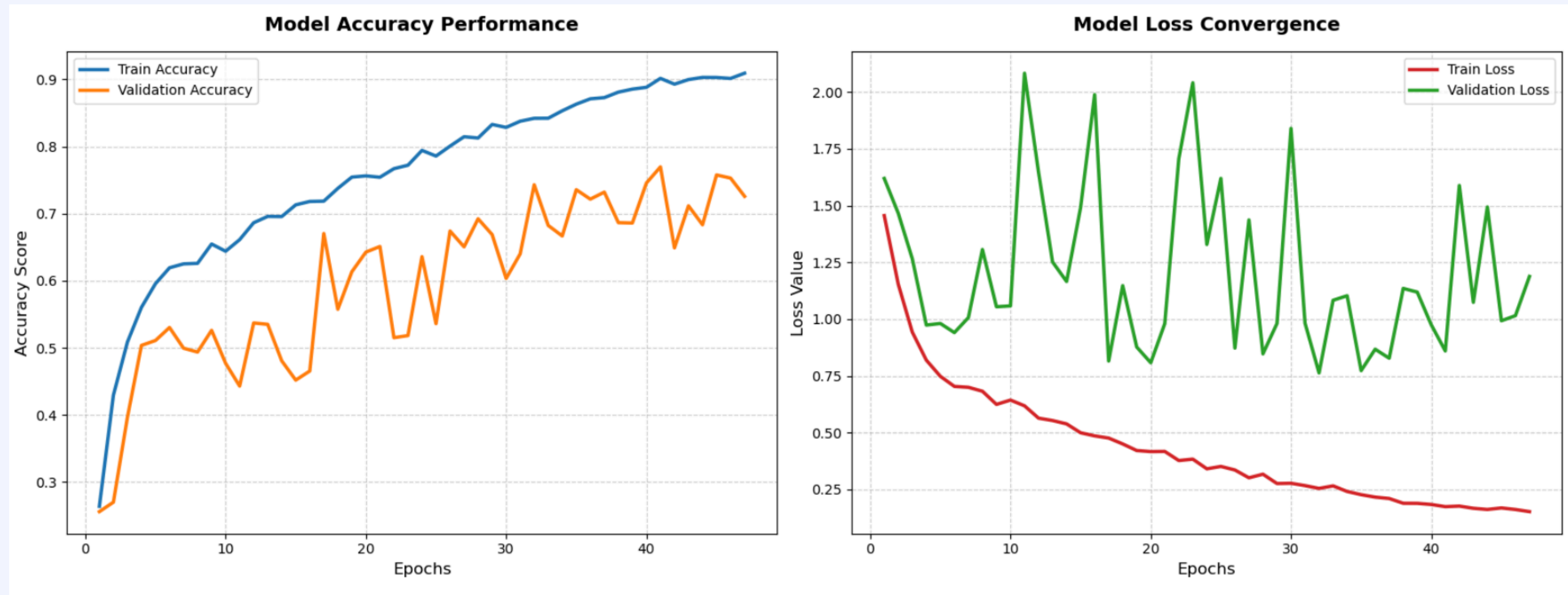
TRAINING & VALIDATION CURVES

1ST VERSION "SIGNALS_EOG1"



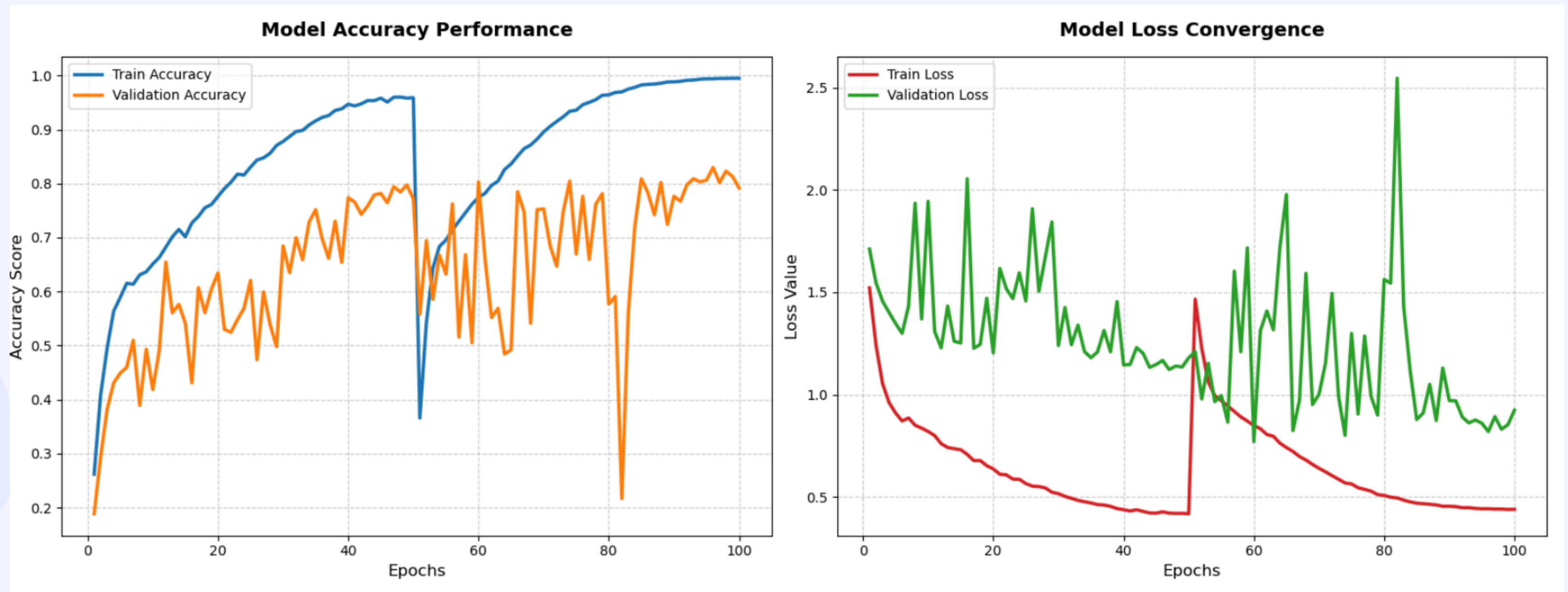
TRAINING & VALIDATION CURVES

2ND VERSION "SIGNALS_EOG2"



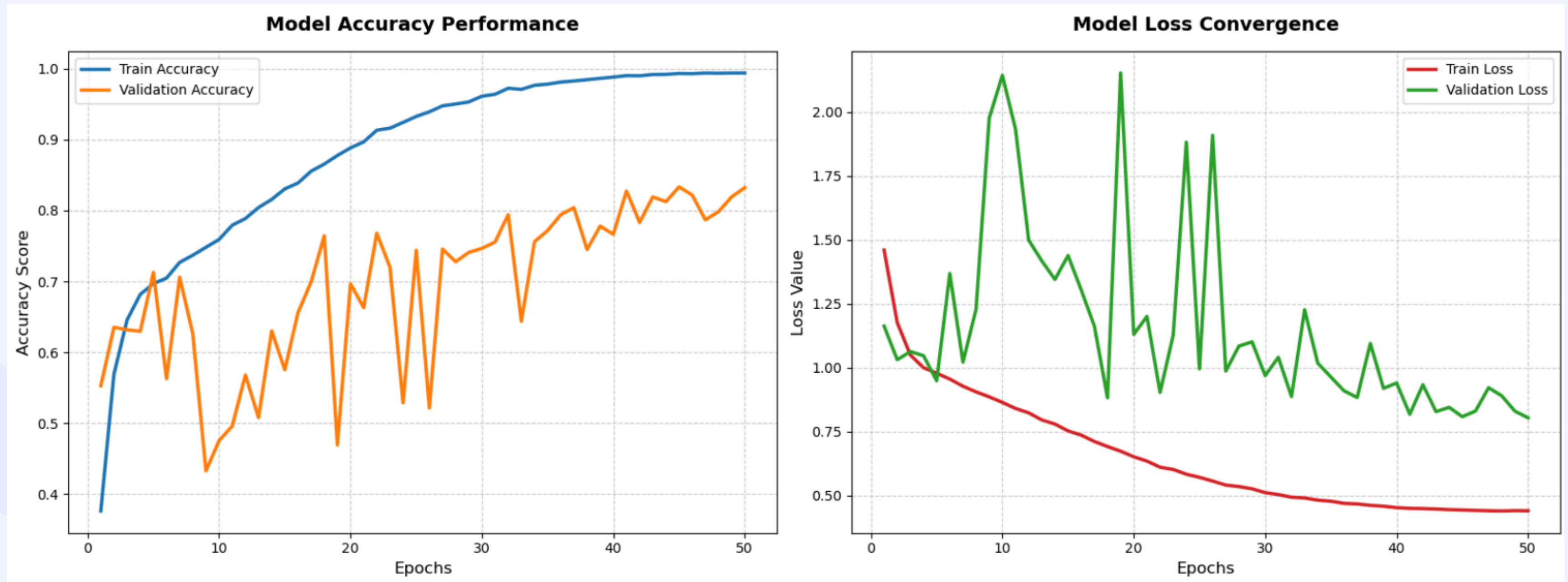
TRAINING & VALIDATION CURVES

3RD VERSION "SIGNALS_EOG3"



TRAINING & VALIDATION CURVES

4TH VERSION "SIGNALS_EOG4"



EVALUATION & RESULTS

1ST VERSION "SIGNALS_EOG1"

Final Test Accuracy: 65.95%

2ND VERSION "SIGNALS_EOG2"

Final Test Accuracy: 74.50%

3RD VERSION "SIGNALS_EOG3"

Final Test Accuracy: 80.25%

4TH VERSION "SIGNALS_EOG4"

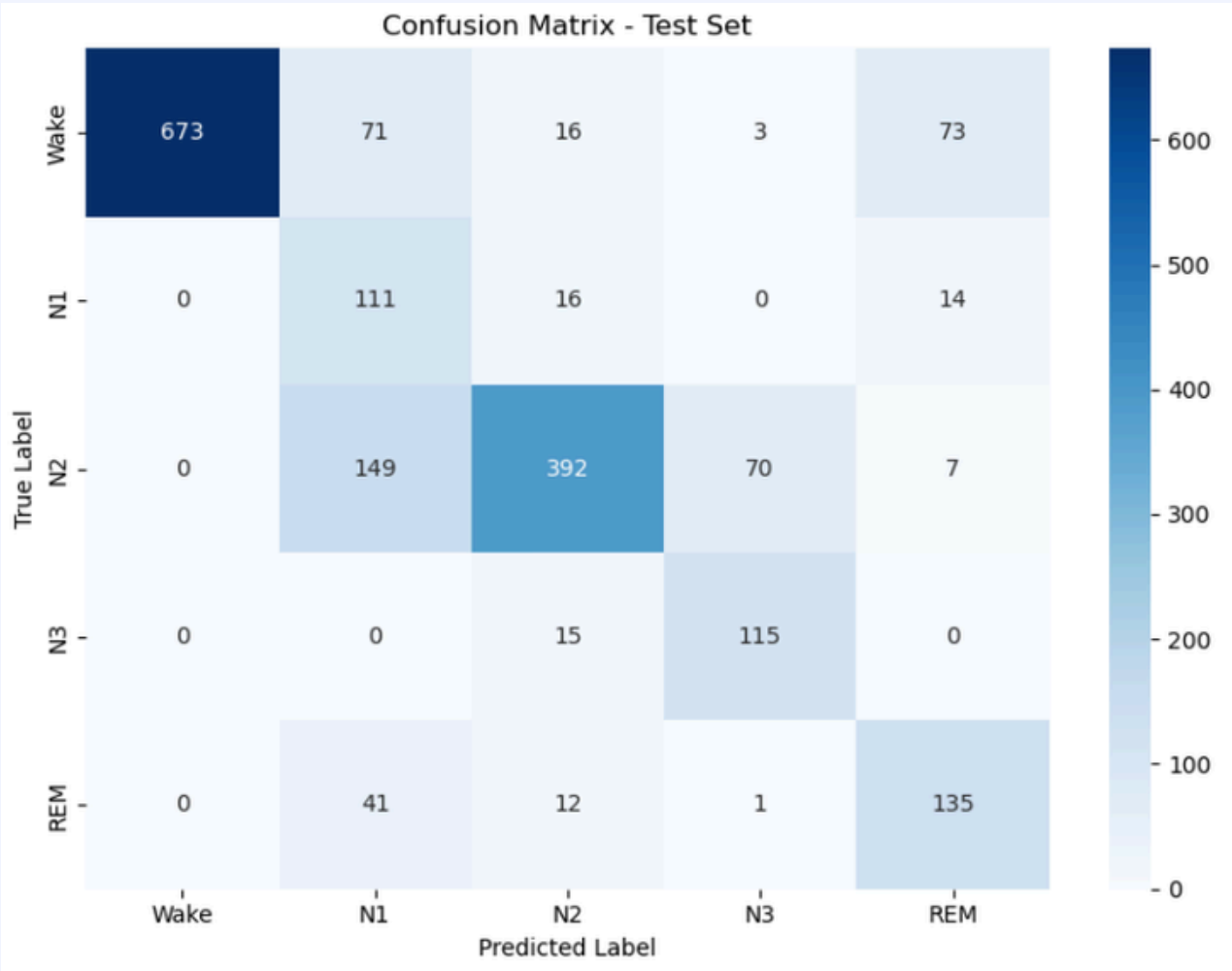
Final Test Accuracy: 82.39%

CONFUSION MATRICES

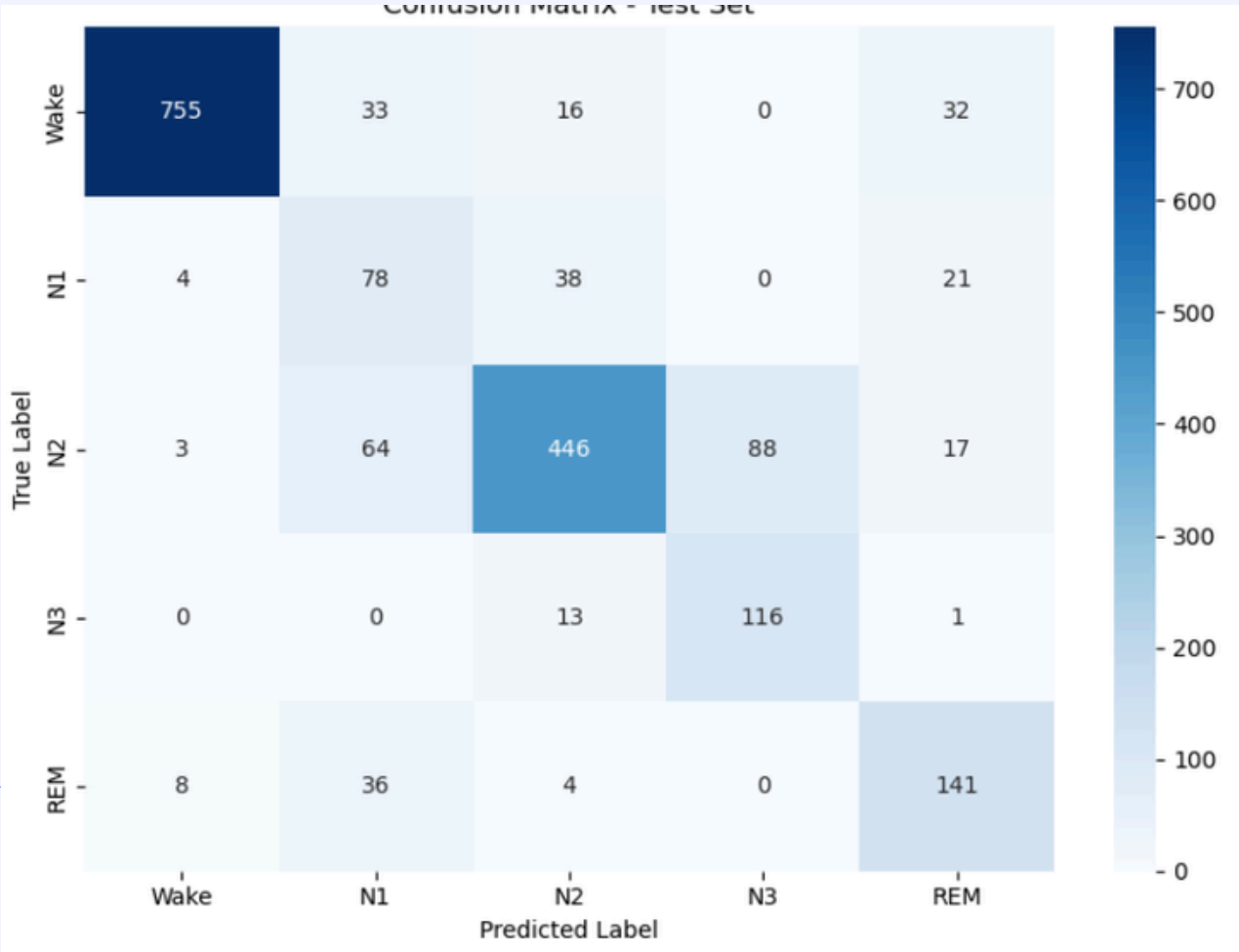
1ST VERSION



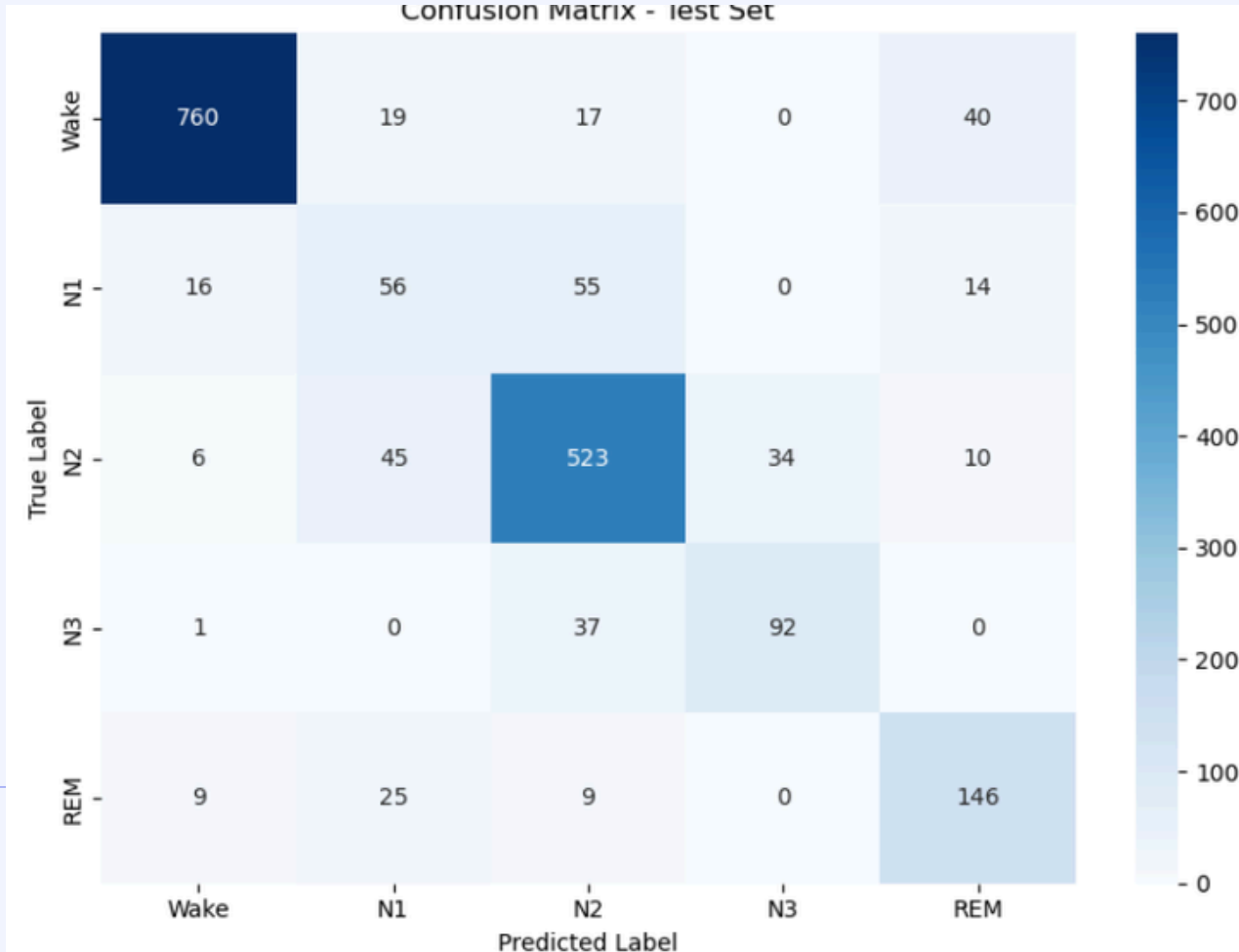
2ND VERSION



3RD VERSION



4TH VERSION



CLASSIFICATION REPORTS

1ST VERSION

	precision	recall	f1-score	support
Wake	0.9987	0.5979	0.7480	1323
N1	0.3214	0.3424	0.3316	184
N2	0.7891	0.7234	0.7548	817
N3	0.5462	0.7429	0.6295	175
REM	0.3116	0.9421	0.4683	259
accuracy			0.6595	2758
macro avg	0.5934	0.6697	0.5864	2758
weighted avg	0.7982	0.6595	0.6884	2758

2ND VERSION

	precision	recall	f1-score	support
Wake	1.0000	0.8050	0.8920	836
N1	0.2984	0.7872	0.4327	141
N2	0.8692	0.6343	0.7334	618
N3	0.6085	0.8846	0.7210	130
REM	0.5895	0.7143	0.6459	189
accuracy			0.7450	1914
macro avg	0.6731	0.7651	0.6850	1914
weighted avg	0.8389	0.7450	0.7710	1914

3RD VERSION

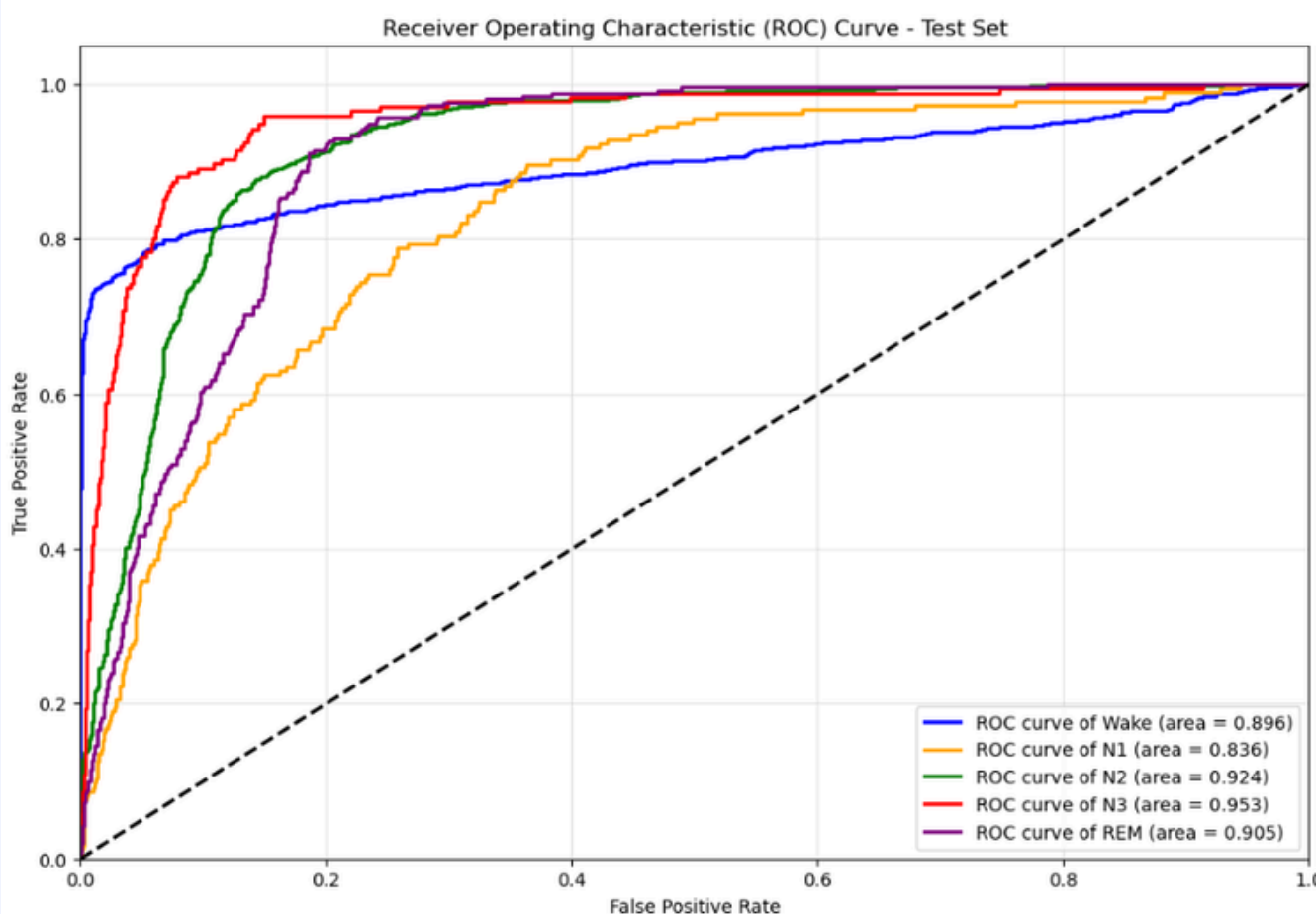
	precision	recall	f1-score	support
Wake	0.9805	0.9031	0.9402	836
N1	0.3697	0.5532	0.4432	141
N2	0.8627	0.7217	0.7859	618
N3	0.5686	0.8923	0.6946	130
REM	0.6651	0.7460	0.7032	189
accuracy			0.8025	1914
macro avg	0.6893	0.7633	0.7134	1914
weighted avg	0.8383	0.8025	0.8137	1914

4TH VERSION

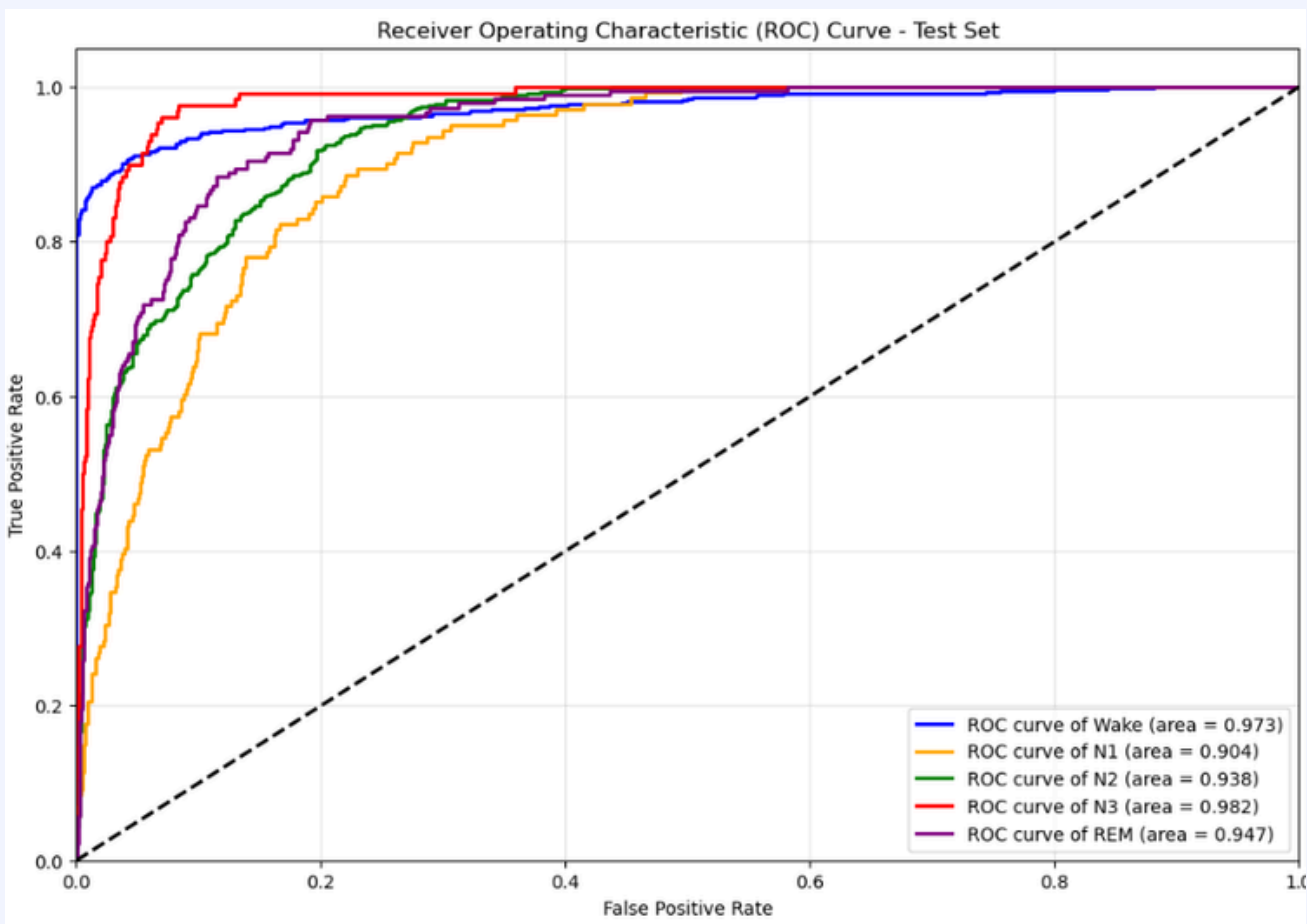
	precision	recall	f1-score	support
Wake	0.9596	0.9091	0.9337	836
N1	0.3862	0.3972	0.3916	141
N2	0.8159	0.8463	0.8308	618
N3	0.7302	0.7077	0.7188	130
REM	0.6952	0.7725	0.7318	189
accuracy			0.8239	1914
macro avg	0.7174	0.7265	0.7213	1914
weighted avg	0.8293	0.8239	0.8260	1914

ROC CURVES

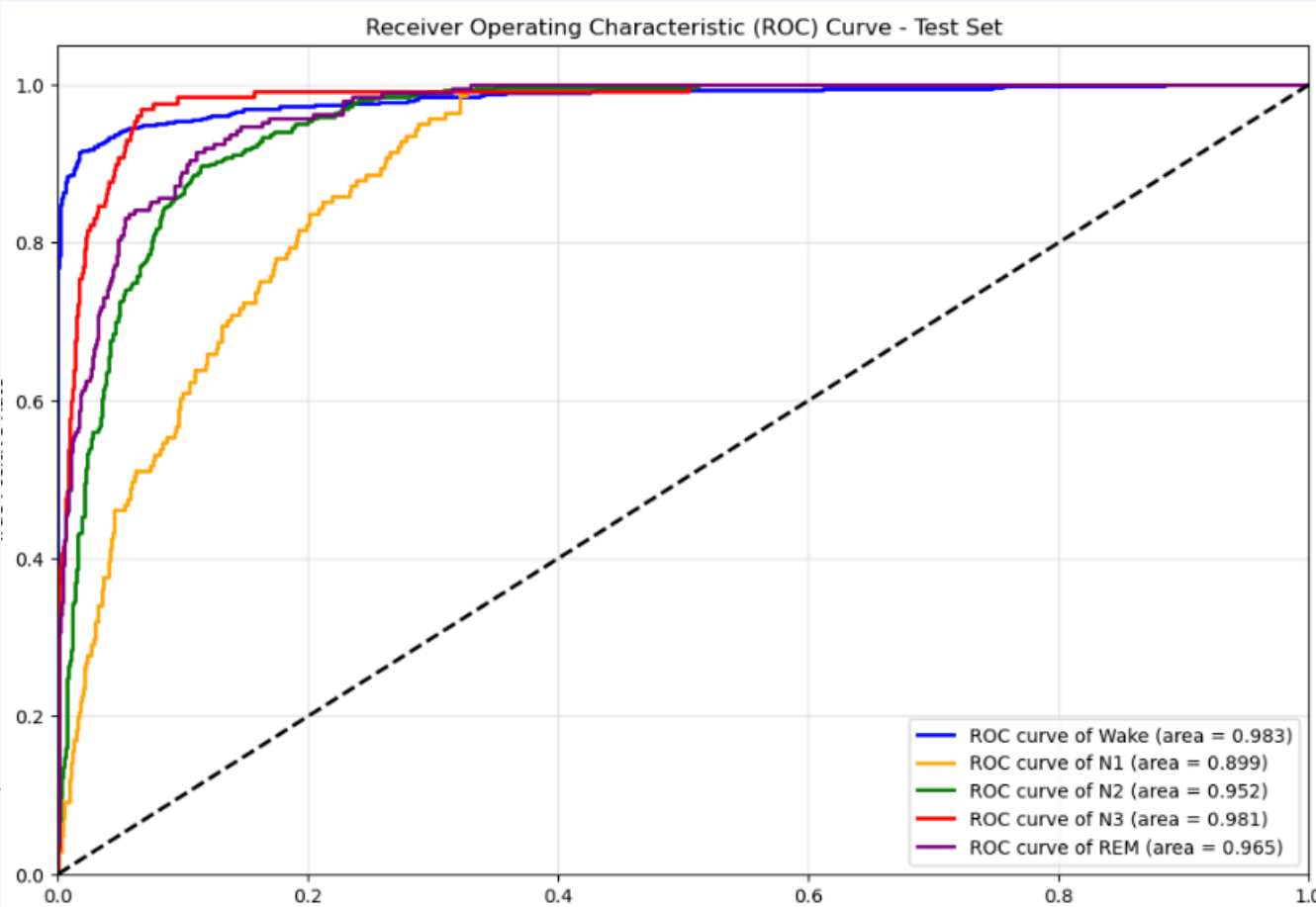
1ST VERSION



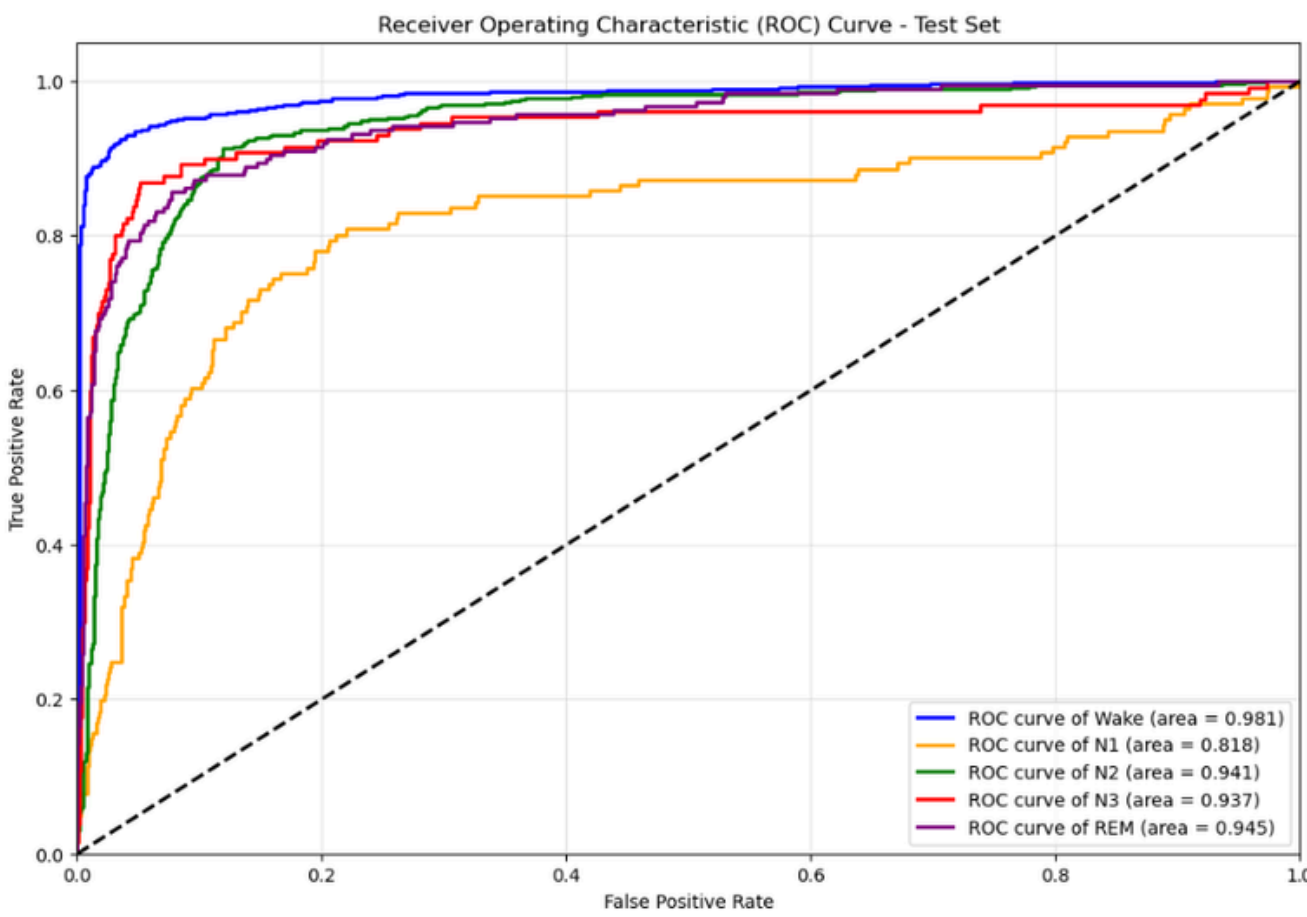
2ND VERSION



3RD VERSION



4TH VERSION



RELATED PAPER

Paper:

U-Sleep: Resilient High-Frequency Sleep Staging

Objective:

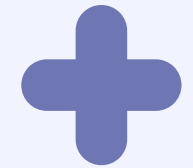
Develop a robust deep learning model for automatic sleep stage classification that works with limited channels (EEG + EOG) across different datasets and clinical settings.

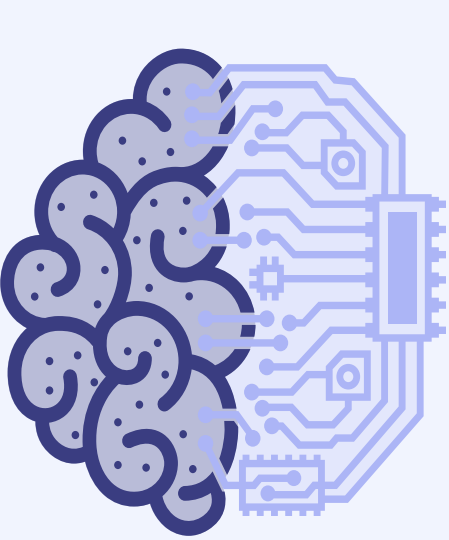
Methodology:

- Uses single EEG + single EOG channels
- Fully Convolutional Neural Network (U-Net inspired architecture)
- Trained on large-scale PSG data including MESA dataset
- Works with different sampling rates and electrode placements
- Classify sleep into five stages: Wake, N1, N2, N3, REM

Results:

- High performance on MESA and multiple other datasets
- Mean F1 score around 0.79 across stages (comparable to human experts)
- Strong results with limited channels (EEG + EOG)
- Very resilient and generalizable to new clinical data





**THANK
YOU**

